

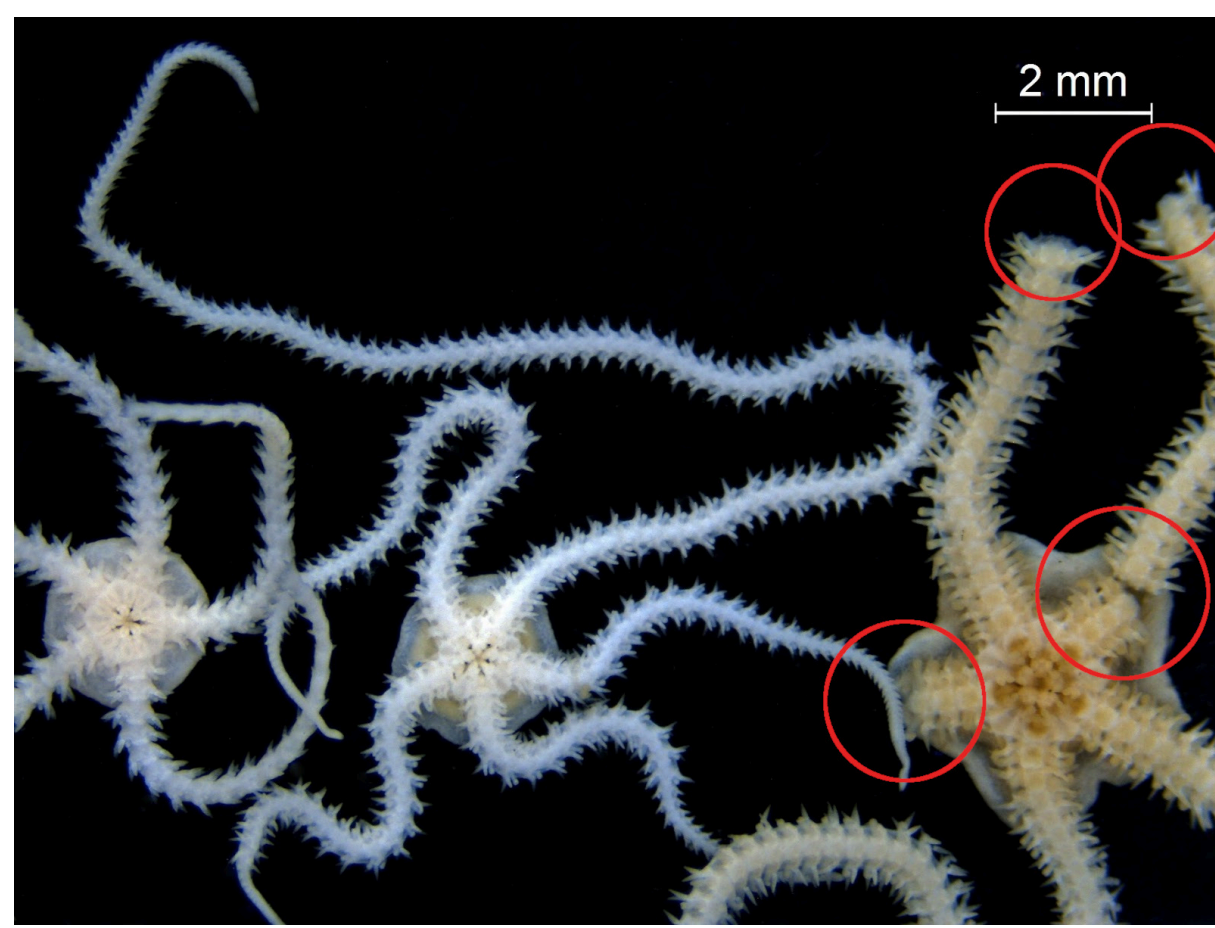
Brittle Star-Inspired Damage Robustness: Multi-Agent Reinforcement Learning Approach

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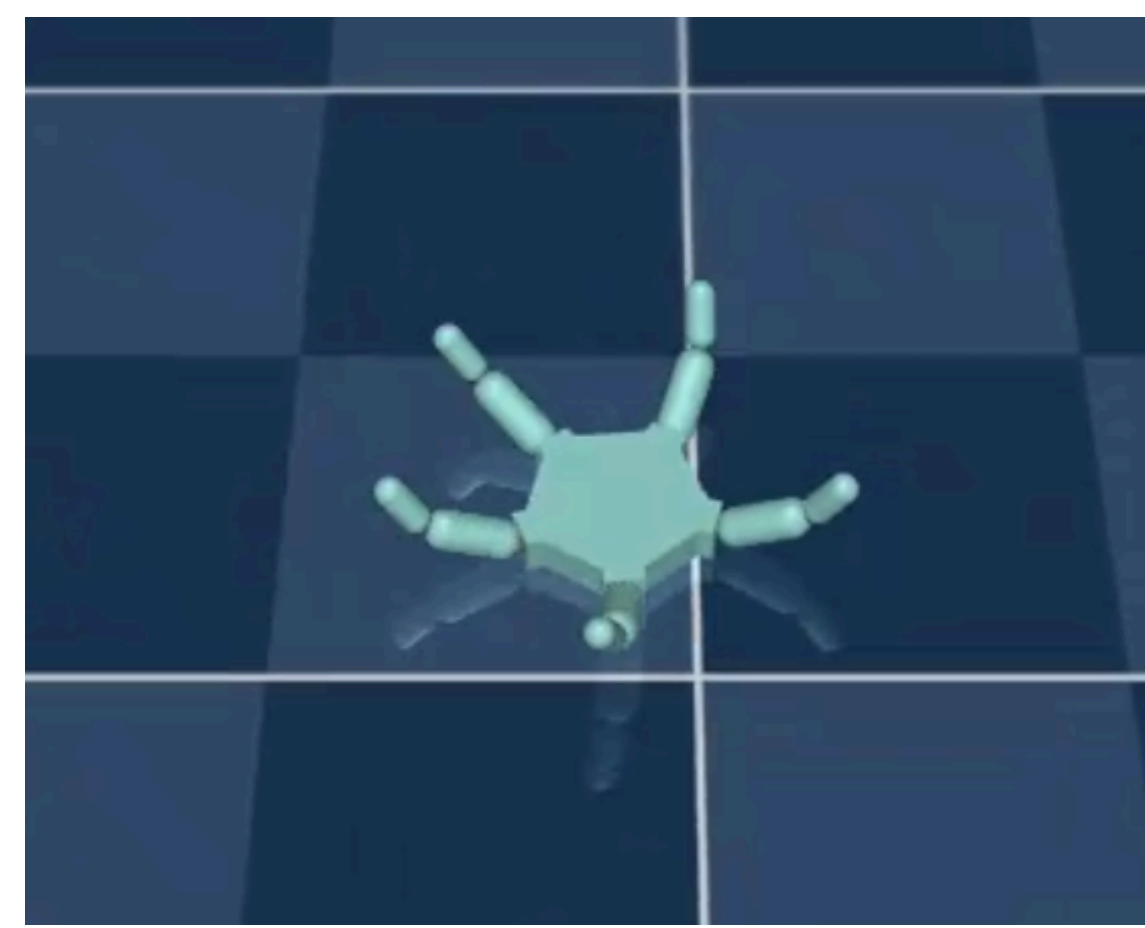
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1 Damage Robustness

Real-world robots might sustain damage in operation. Nature already has solutions to this problem, e.g. brittle stars:



Two intact (left) and one damaged (right) brittle stars.



Brittle star-inspired robot with 5 arms in a simulated environment.

brittle star damage robustness $\rightarrow$ damage-robust robots

2 Research Goal

Build damage-robust robots by learning brittle star-inspired locomotion gaits with the help of central pattern generators (CPGs) and multi-agent reinforcement learning (MARL).

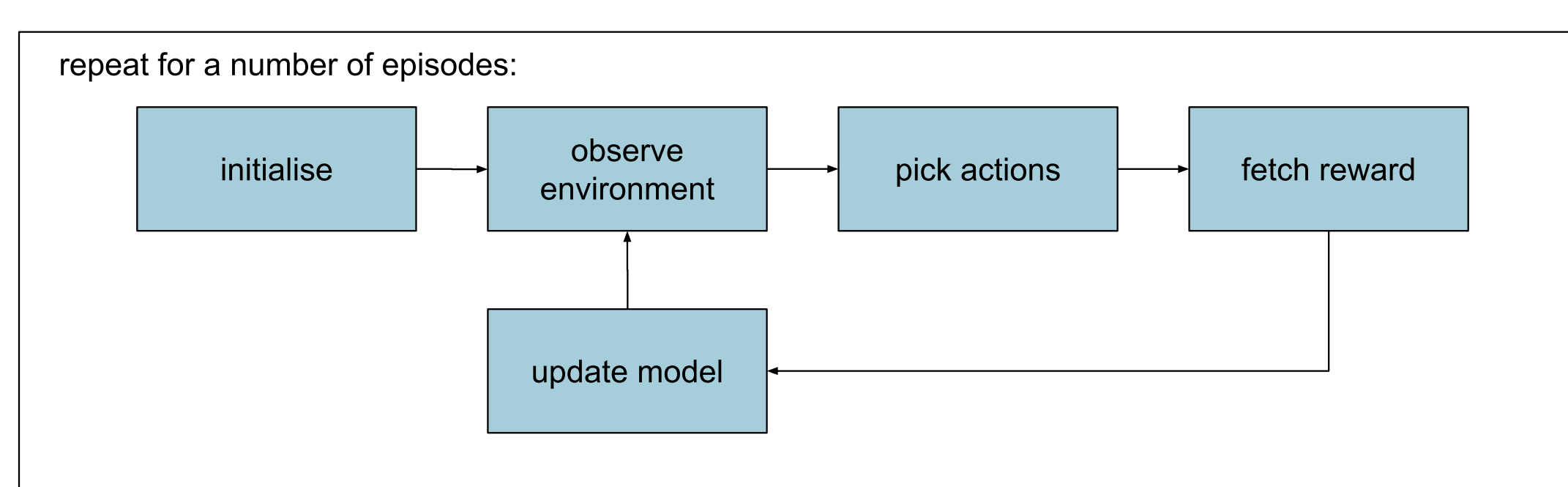
3 Methodology

Actions: rowing gait modulation through CPGs:

Action	Description
Leading arm	Pointing upward
Left rower	Clockwise rotation
Right rower	Counterclockwise rotation
Secondary left rower	Smaller amplitude clockwise
Secondary right rower	Smaller amplitude counterclockwise

Learning: **action selection** through MARL:

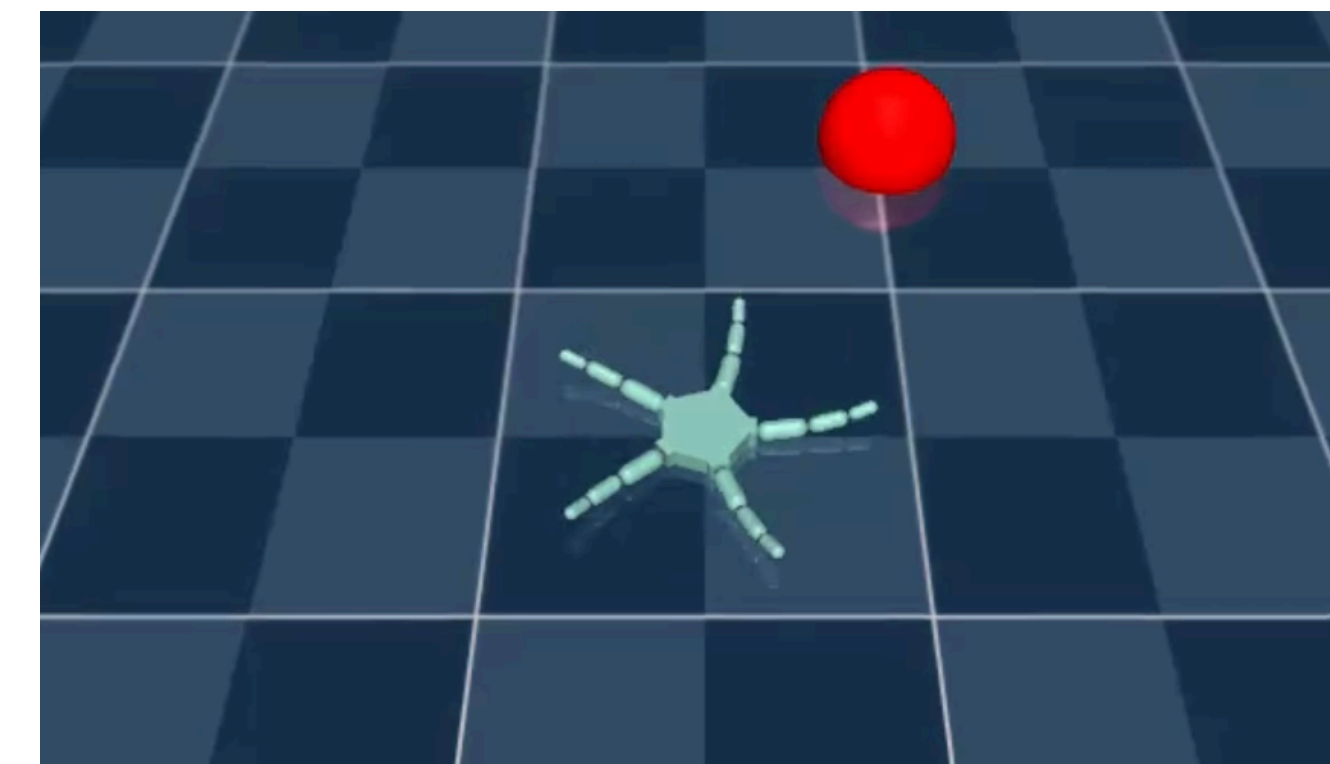
- Observations:
 1. **Angle** between arm and target,
 2. **Direction** between body and target,
 3. **Distance** between body and target
- Reward function: **distance reduction** toward target
- **Fixed** target position



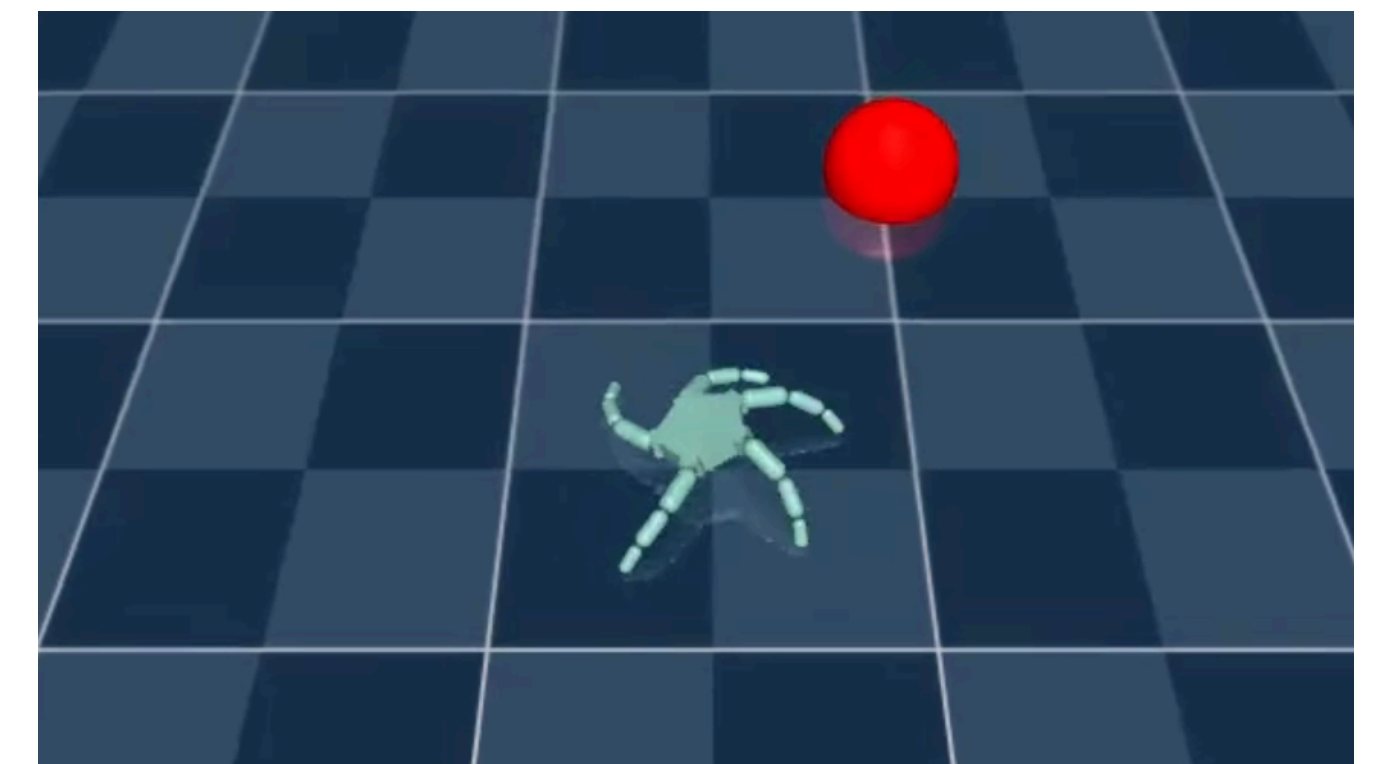
Overview of the methodology.

4 Results

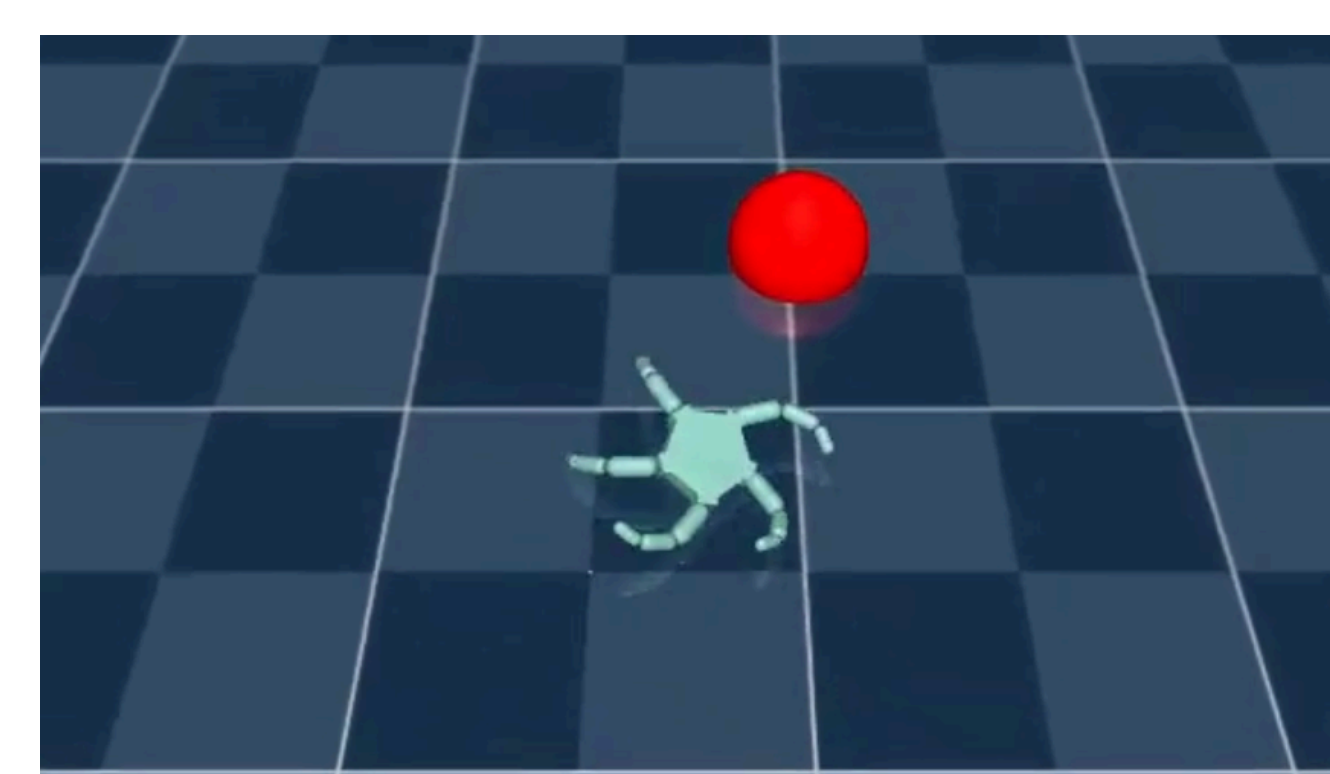
Can walk in simulation with 5 arms, fixed target and no damage:



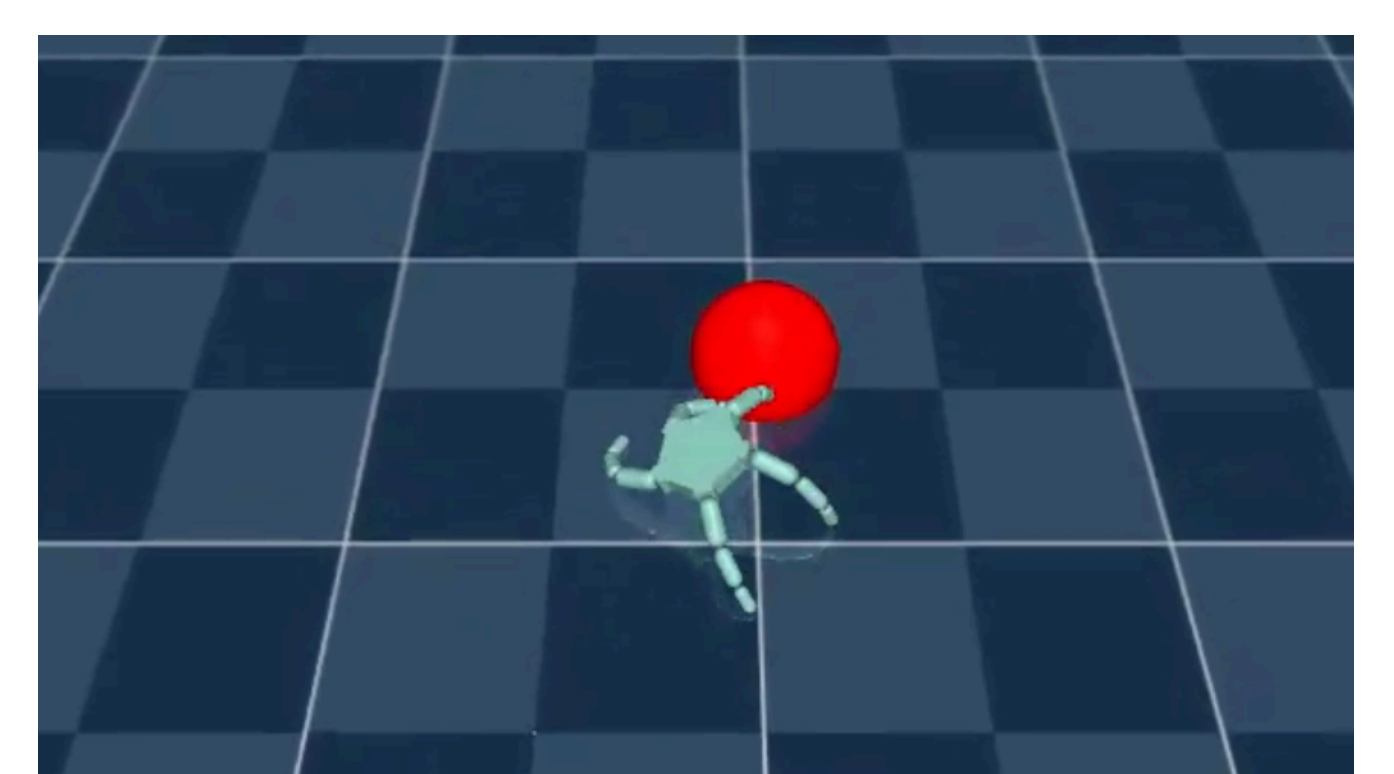
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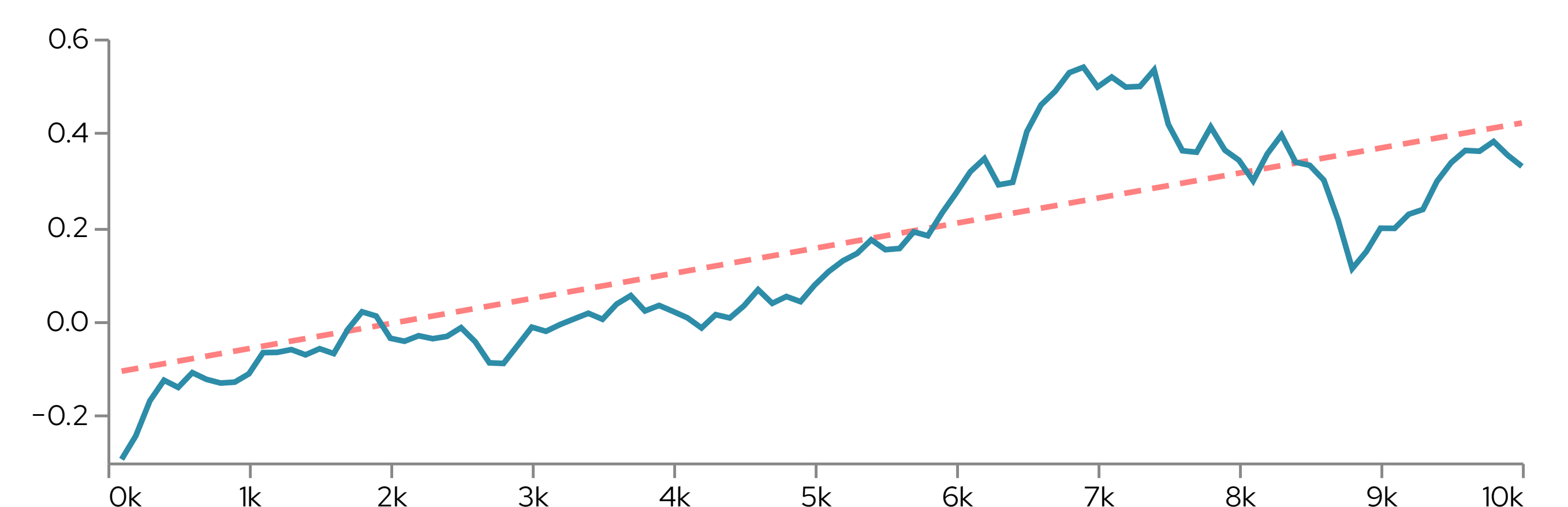


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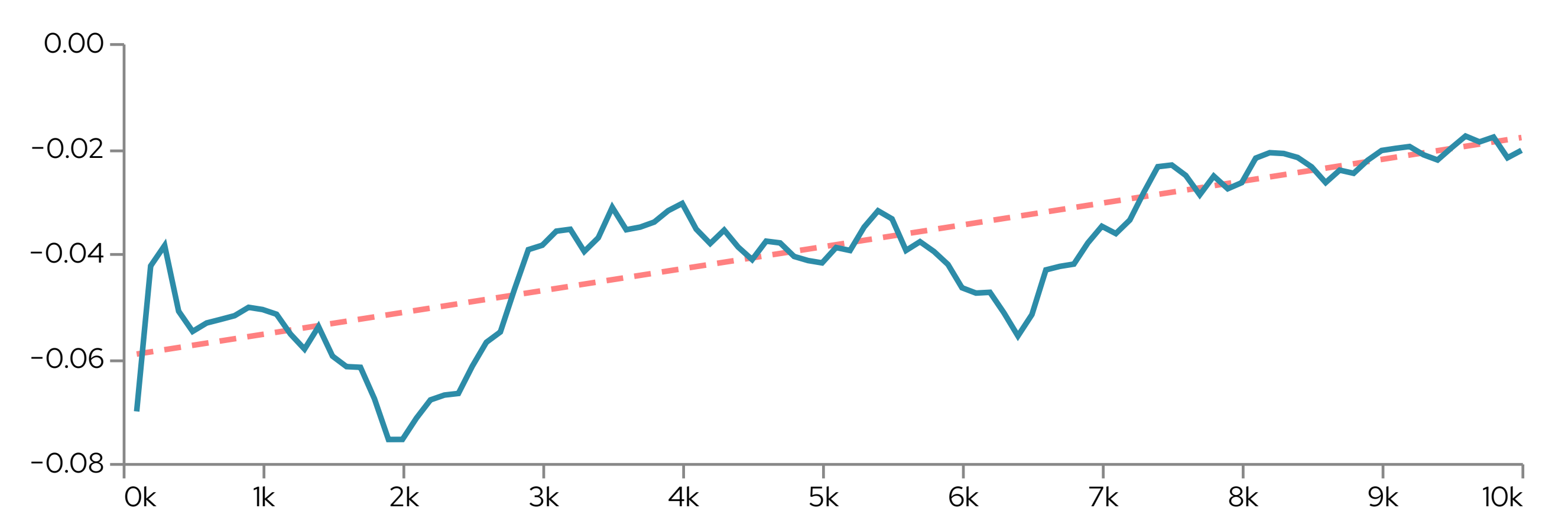
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Reward functions shows a **small increase** in reward, but remains close to zero:



Average reward over 10 environment steps while training for 100 episodes

Moving to random target positions shows the **same results, but with a lower average** reward:



Same training setup as above, but with random target positions

5 Conclusions

1. More research is required to **fix the learning process**.
2. Reward function should be checked for **bugs**.
3. Cannot say anything about **damage robustness**